

GLOBAL
EDITION



Statistics for Managers

Using Microsoft® Excel®

NINTH EDITION

David M. Levine • David F. Stephan • Kathryn A. Szabat



Chapter 2

Organising and Visualising Variables

Objectives

In this chapter you learn:

- How to organise and visualise categorical variables.
- How to organise and visualise numerical variables.
- How to summarise a mix of variables.
- How to avoid making common errors when organising and visualising variables.

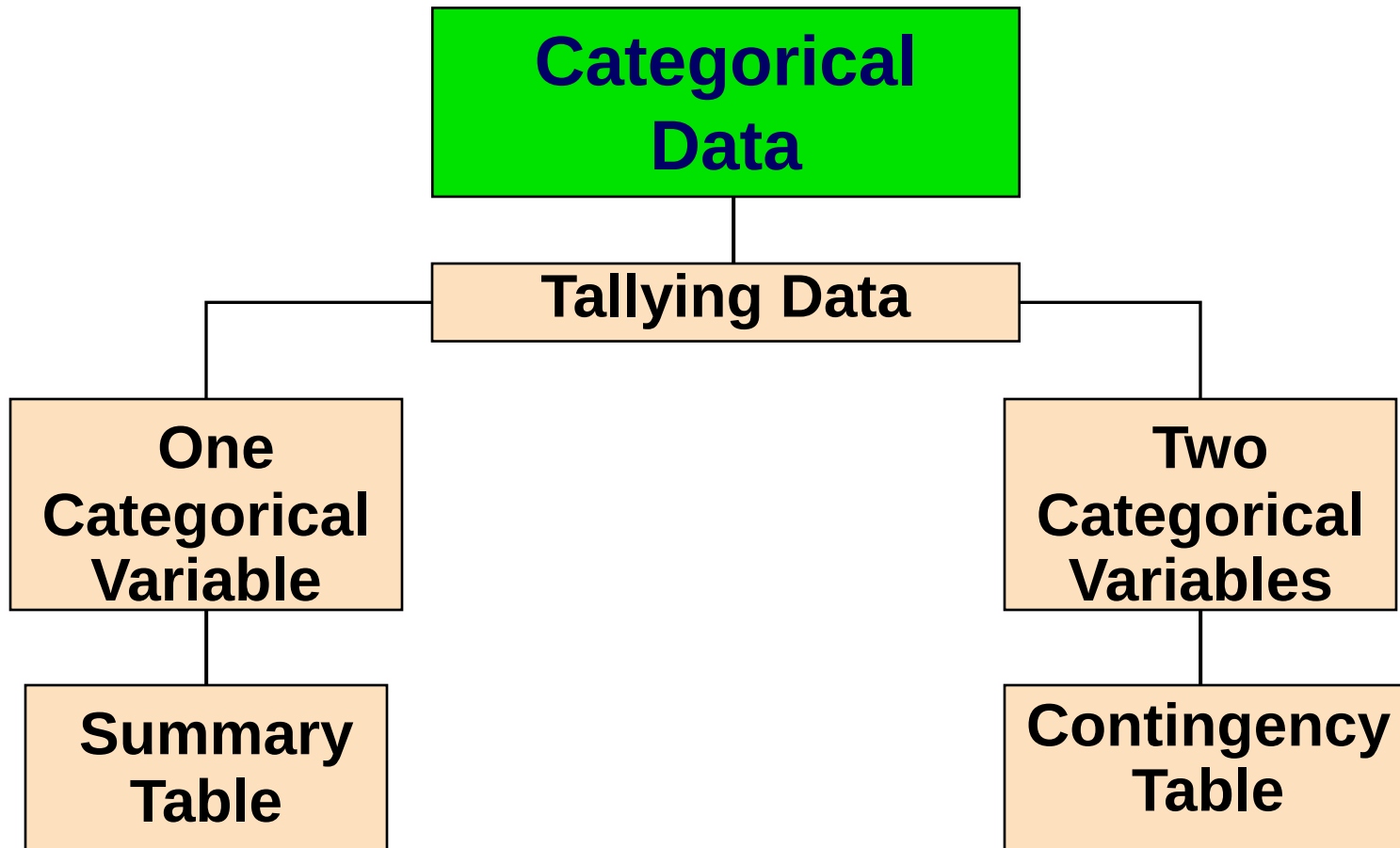
Organising Data Creates Both Tabular And Visual Summaries

DCOVA

- Summaries both guide further exploration and sometimes facilitate decision making.
- Visual summaries enable rapid review of larger amounts of data & show possible significant patterns.
- Often, the **Organise** and **Visualise** step in **DCOVA** occur concurrently.

Categorical Data Are Organised By Utilising Tables

DCOVA



Organising Categorical Data: Summary Table

DCOVA

- A **summary table** tallies the frequencies or percentages of items in a set of categories so that you can see differences between categories.

Devices Millennials Use to Watch Movies or Television Shows

Devices Used To Watch Movies or TV Shows	Percent
Television Set	49%
Tablet	9%
Smartphone	10%
Laptop / Desktop	32%

Source: Data extracted and adapted from A. Sharma, "Big Media Needs to Embrace Digital Shift Not Fight It," Wall Street Journal, June 22, 2016, p. 1-2.

A Contingency Table Helps Organise Two or More Categorical Variables

DCOVA

- Used to study patterns that may exist between the responses of two or more categorical variables.
- Cross tabulates or tallies jointly the responses of the categorical variables.
- For two variables the tallies for one variable are in the rows and the tallies for the second variable are in the columns.

Contingency Table - Example

DCOVA

- A random sample of 400 invoices is drawn.
- Each invoice is categorised as a small, medium, or large amount.
- Each invoice is also examined to identify if there are any errors.
- This data are then organised in the contingency table to the right.

Contingency Table Showing Frequency of Invoices Categorised By Size and The Presence Of Errors

	No Errors	Errors	Total
Small Amount	170	20	190
Medium Amount	100	40	140
Large Amount	65	5	70
Total	335	65	400

Contingency Table Based On Percentage Of Overall Total

DCOVA

	No Errors	Errors	Total
Small Amount	170	20	190
Medium Amount	100	40	140
Large Amount	65	5	70
Total	335	65	400

$42.50\% = 170 / 400$
 $25.00\% = 100 / 400$
 $16.25\% = 65 / 400$

	No Errors	Errors	Total
Small Amount	42.50%	5.00%	47.50%
Medium Amount	25.00%	10.00%	35.00%
Large Amount	16.25%	1.25%	17.50%
Total	83.75%	16.25%	100.0%

83.75% of sampled invoices have no errors and 47.50% of sampled invoices are for small amounts.

Contingency Table Based On Percentage of Row Totals

DCOVA

	No Errors	Errors	Total
Small Amount	170	20	190
Medium Amount	100	40	140
Large Amount	65	5	70
Total	335	65	400

$89.47\% = 170 / 190$
 $71.43\% = 100 / 140$
 $92.86\% = 65 / 70$

	No Errors	Errors	Total
Small Amount	89.47%	10.53%	100.0%
Medium Amount	71.43%	28.57%	100.0%
Large Amount	92.86%	7.14%	100.0%
Total	83.75%	16.25%	100.0%

Medium invoices have a larger chance (28.57%) of having errors than small (10.53%) or large (7.14%) invoices.

Contingency Table Based On Percentage Of Column Totals

DCoVA

	No Errors	Errors	Total
Small Amount	170	20	190
Medium Amount	100	40	140
Large Amount	65	5	70
Total	335	65	400

$$50.75\% = 170 / 335$$

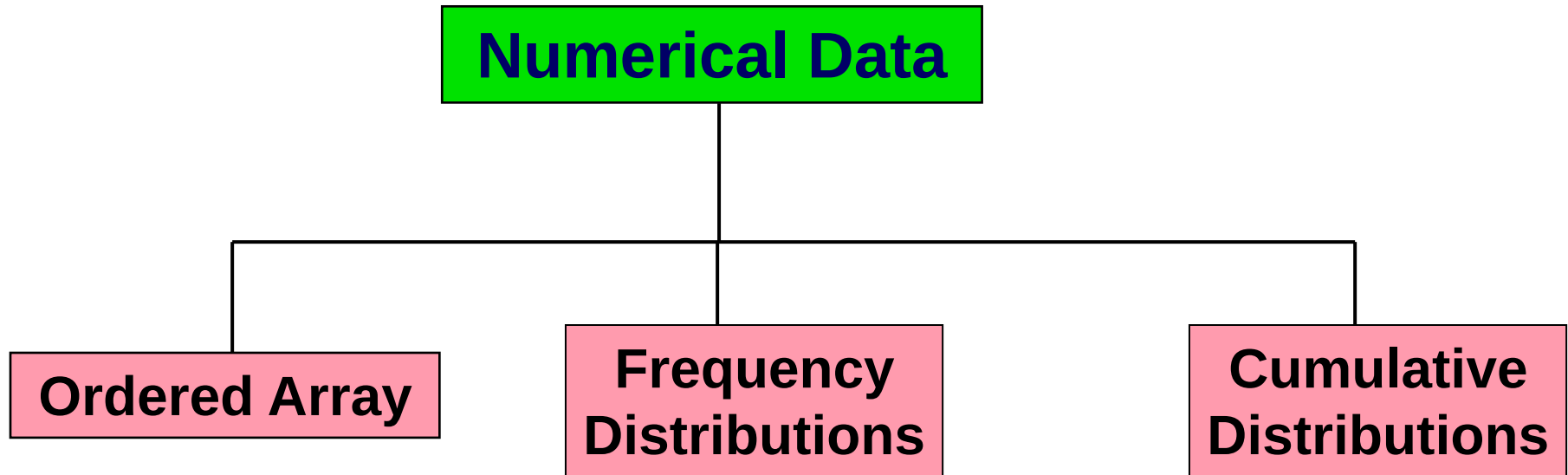
$$30.77\% = 20 / 65$$

There is a 61.54% chance that invoices with errors are of medium size.

	No Errors	Errors	Total
Small Amount	50.75%	30.77%	47.50%
Medium Amount	29.85%	61.54%	35.00%
Large Amount	19.40%	7.69%	17.50%
Total	100.0%	100.0%	100.0%

Tables Used For Organising Numerical Data

DCOVA



Organising Numerical Data: Ordered Array

DCOVA

- An **ordered array** is a sequence of data, in rank order, from the smallest value to the largest value.
- Shows range (minimum value to maximum value).
- May help identify outliers (unusual observations).

Age of Surveyed College Students	Day Students					
	16	17	17	18	18	18
	19	19	20	20	21	22
	22	25	27	32	38	42
	Night Students					
	18	18	19	19	20	21
	23	28	32	33	41	45

Organising Numerical Data: Frequency Distribution

DCQVA

- The **frequency distribution** is a summary table in which the data are arranged into numerically ordered classes.
- You must give attention to selecting the appropriate *number* of **class groupings** for the table, determining a suitable *width* of a class grouping, and establishing the *boundaries* of each class grouping to avoid overlapping.
- The number of classes depends on the number of values in the data. With a larger number of values, typically there are more classes. In general, a frequency distribution should have at least 5 but no more than 15 classes.
- To determine the **width of a class interval**, you divide the **range** (Highest value–Lowest value) of the data by the number of class groupings desired.

Organising Numerical Data: Frequency Distribution Example

DCOVA

Example: A manufacturer of insulation randomly selects 20 winter days and records the daily high temperature in degrees Fahrenheit.

24, 35, 17, 21, 24, 37, 26, 46, 58, 30, 32, 13, 12, 38, 41, 43, 44, 27, 53, 27

Organising Numerical Data: Frequency Distribution Example

DCOVA

- Sort raw data in ascending order:
12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58.
- Find range: **$58 - 12 = 46$.**
- Select number of classes: **5 (usually between 5 and 15).**
- Compute class interval (width): **10 ($46/5$ then round up).**
- Determine class boundaries (limits):
 - Class 1: 10 but less than 20.
 - Class 2: 20 but less than 30.
 - Class 3: 30 but less than 40.
 - Class 4: 40 but less than 50.
 - Class 5: 50 but less than 60.
- Compute class midpoints: **15, 25, 35, 45, 55.**
- Count observations & assign to classes.

Organising Numerical Data: Frequency Distribution Example

DCOVA

Data in ordered array:

12, 13, 17, 21, 24, 24, 26, 27, 27, 30, 32, 35, 37, 38, 41, 43, 44, 46, 53, 58

Class	Midpoints	Frequency
10 but less than 20	15	3
20 but less than 30	25	6
30 but less than 40	35	5
40 but less than 50	45	4
50 but less than 60	55	2
Total		20

Organising Numerical Data: Relative & Percent Frequency Distribution Example

DCQVA

Class	Frequency	Relative Frequency	Percentage
10 but less than 20	3	.15	15%
20 but less than 30	6	.30	30%
30 but less than 40	5	.25	25%
40 but less than 50	4	.20	20%
50 but less than 60	2	.10	10%
Total	20	1.00	100%

Relative Frequency = Frequency / Total,

e.g. $0.10 = 2 / 20$

Organising Numerical Data: Cumulative Frequency Distribution Example

DCQVA

Class	Frequency	Percentage	Cumulative Frequency	Cumulative Percentage
10 but less than 20	3	15%	3	15%
20 but less than 30	6	30%	9	45%
30 but less than 40	5	25%	14	70%
40 but less than 50	4	20%	18	90%
50 but less than 60	2	10%	20	100%
Total	20	100%	20	100%

Cumulative Percentage = Cumulative Frequency / Total * 100

e.g. 45% = 100*9/20

Addendum on Notation for Frequency Distributions

- To accommodate continuous variables, the square and rounded bracket notion is used.
- A square bracket denotes inclusive while a rounded bracket denotes exclusive.
- For Example:

Class	Bracket Notation	Description	Examples
1	(20; 30]	Greater than 20 and smaller or equal to 30	Would not include 20, but would include values such as 20.0001. Would include 30, but not 30.100
2	(30; 40]	Greater than 30 and smaller or equal to 40	Would not include 30, but would include values such as 30.001

Why Use a Frequency Distribution?

DCOVA

- It condenses the raw data into a more useful form.
- It allows for a quick visual interpretation of the data.
- It enables the determination of the major characteristics of the data set including where the data are concentrated / clustered.

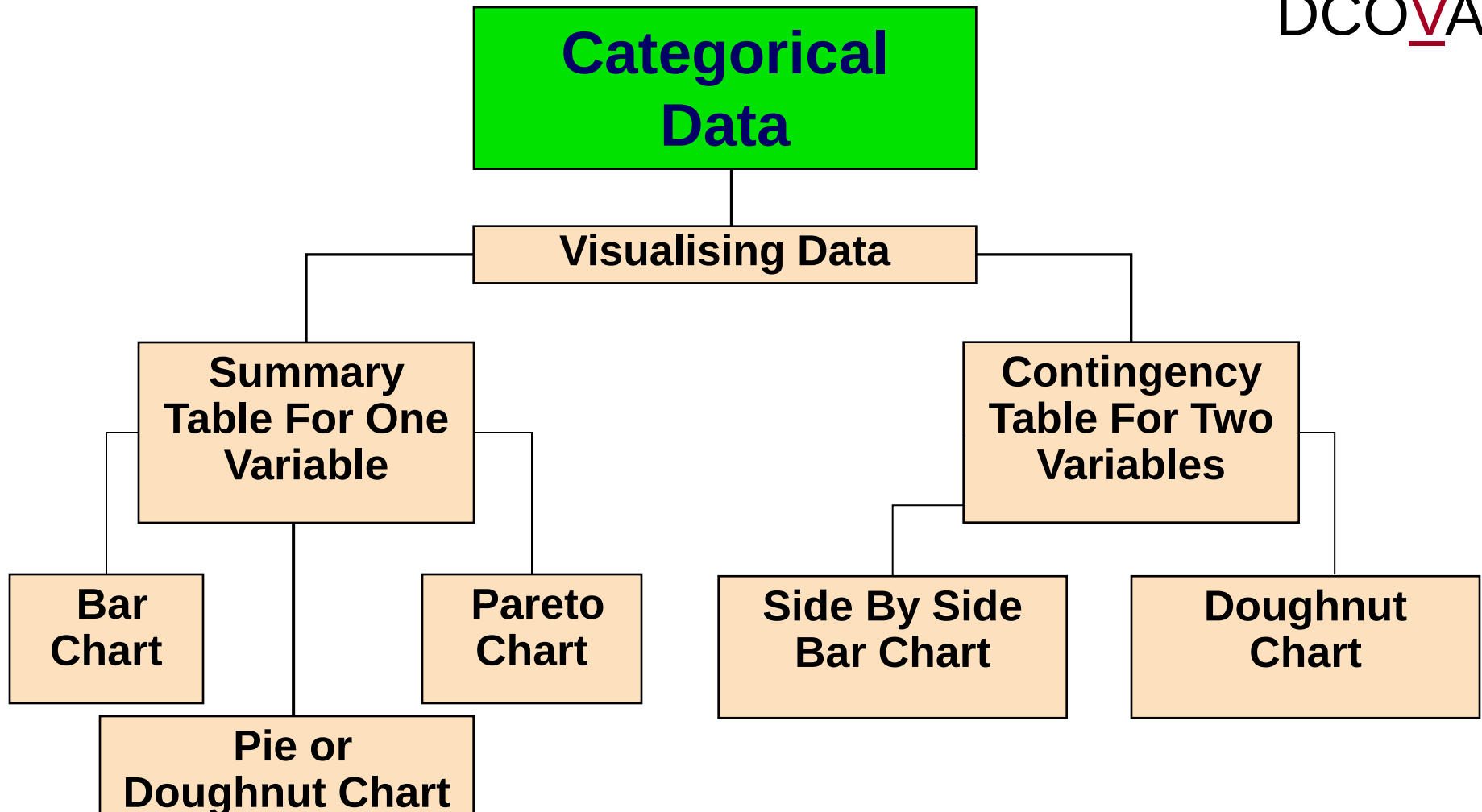
Frequency Distributions: Some Tips

DCOVA

- Different class boundaries may provide different pictures for the same data (especially for smaller data sets).
- Shifts in data concentration may show up when different class boundaries are chosen.
- As the size of the data set increases, the impact of alterations in the selection of class boundaries is greatly reduced.
- When comparing two or more groups with different sample sizes, you must use either a relative frequency or a percentage distribution.

Visualising Categorical Data Through Graphical Displays

DCOVA

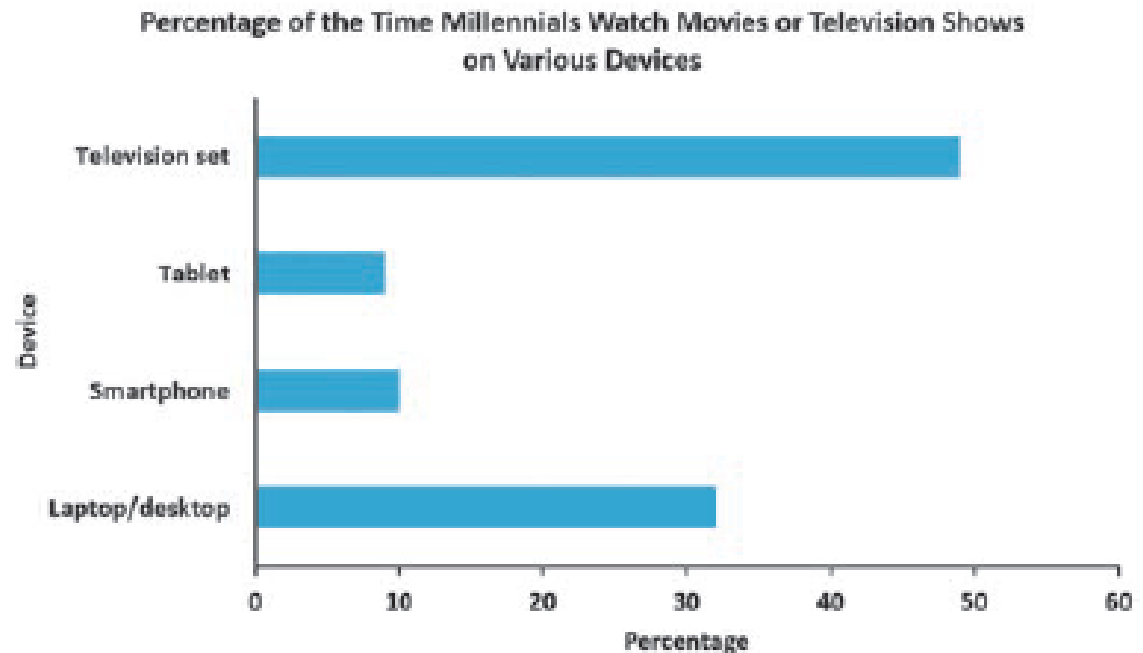


Visualising Categorical Data: The Bar Chart

DCOVA

- The **bar chart** visualises a categorical variable as a series of bars. The length of each bar represents either the frequency or percentage of values for each category. Each bar is separated by a space called a gap.

Devices Used to Watch	Percent
Television Set	49%
Tablet	9%
Smartphone	10%
Laptop / Desktop	32%



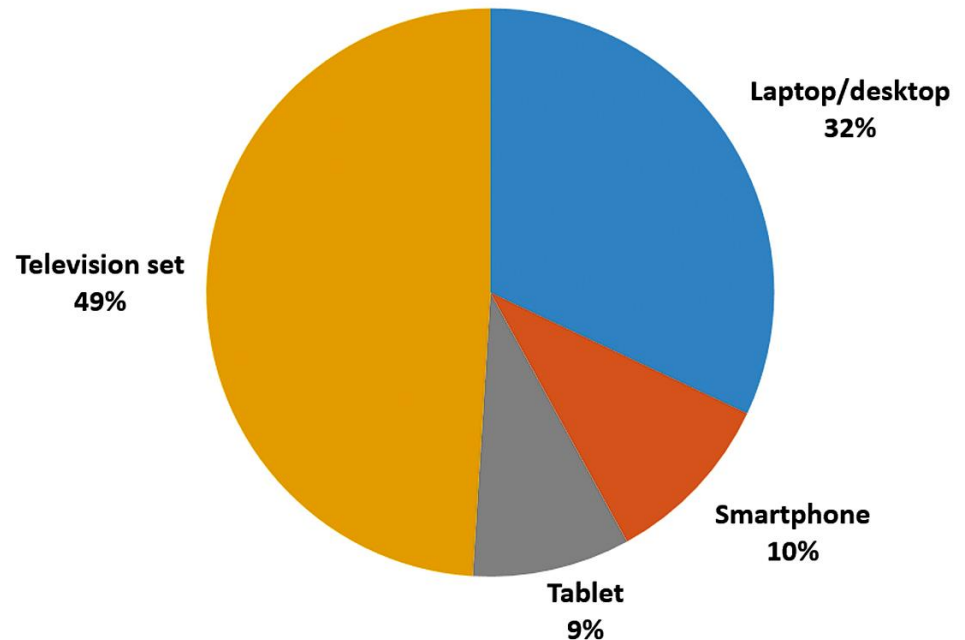
Visualising Categorical Data: The Pie Chart

DCOVA

- The **pie chart** is a circle broken up into slices that represent categories. The size of each slice of the pie varies according to the percentage in each category.

Percentage of the Time Millennials Watch Movies or Television Shows on Various Devices

Devices Used to Watch	Percent
Television Set	49%
Tablet	9%
Smartphone	10%
Laptop / Desktop	32%

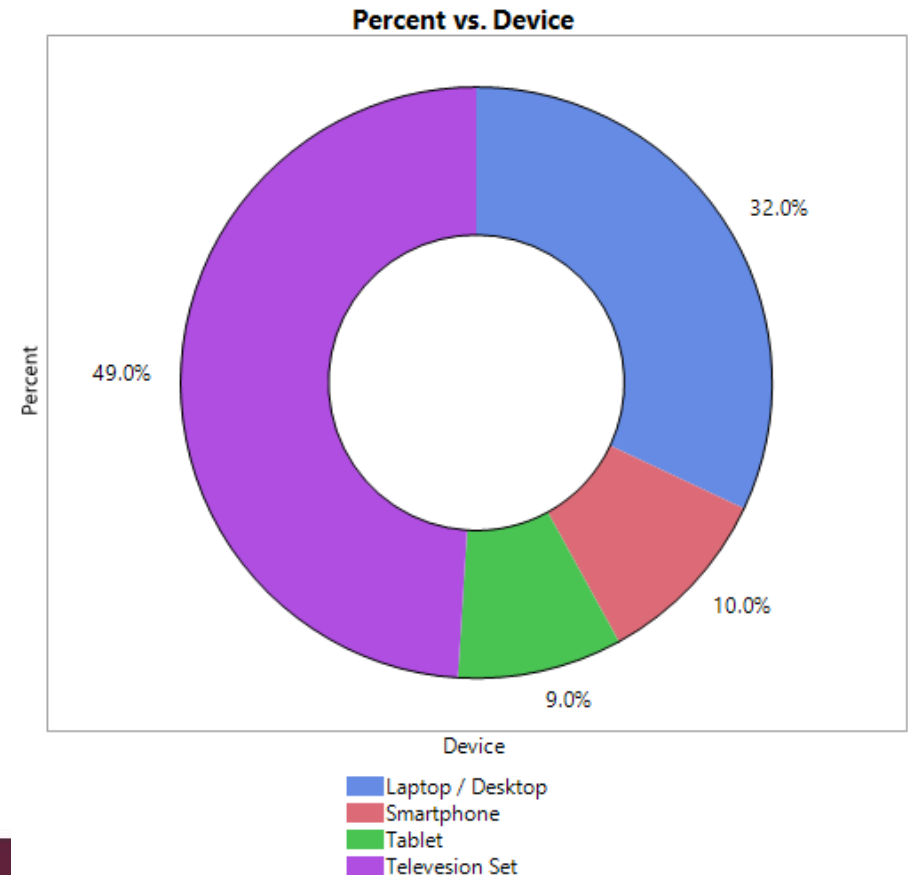


Visualising Categorical Data: The Doughnut Chart

DCOVA

- The **doughnut chart** is the outer part of a circle broken up into pieces that represent categories. The size of each piece of the doughnut varies according to the percentage in each category.

Devices Used to Watch	Percent
Television Set	49%
Tablet	9%
Smartphone	10%
Laptop / Desktop	32%



Visualising Categorical Data: The Pareto Chart

DCOVA

- Used to portray categorical data (nominal scale).
- A vertical bar chart, where categories are shown in descending order of frequency.
- A cumulative polygon is shown in the same graph.
- Used to separate the “vital few” from the “trivial many.”

Visualising Categorical Data: The Pareto Chart (con't)

DCOVA

Ordered Summary Table For Causes Of Incomplete ATM Transactions

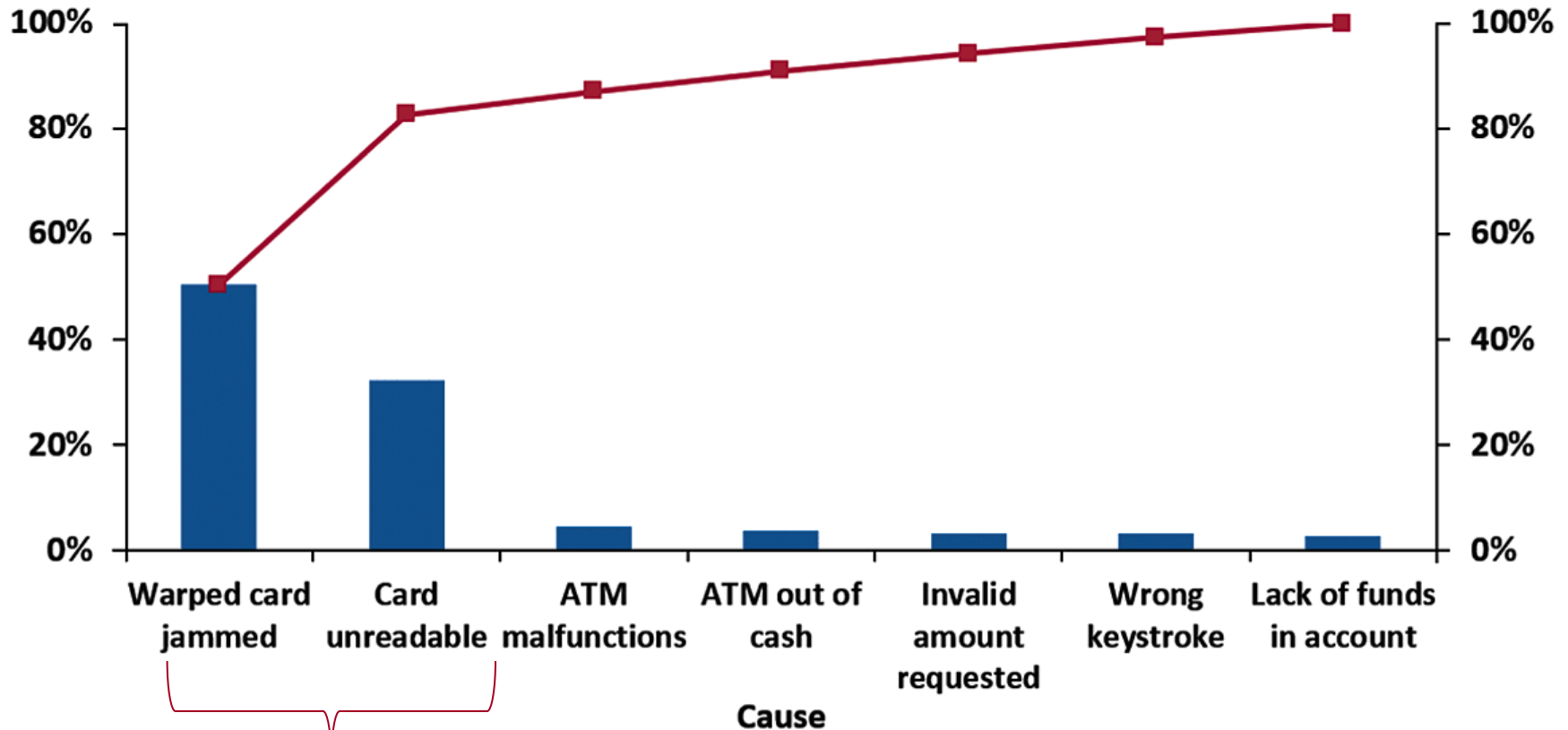
Cause	Frequency	Percent	Cumulative Percent
Warped card jammed	365	50.41%	50.41%
Card unreadable	234	32.32%	82.73%
ATM malfunctions	32	4.42%	87.15%
ATM out of cash	28	3.87%	91.02%
Invalid amount requested	23	3.18%	94.20%
Wrong keystroke	23	3.18%	97.38%
Lack of funds in account	19	2.62%	100.00%
Total	724	100.00%	

Source: Data extracted from A. Bhalla, "Don't Misuse the Pareto Principle," *Six Sigma Forum Magazine*, May 2009, pp. 15–18.

Visualising Categorical Data: The Pareto Chart (con't)

DCOVA

Pareto Chart of Incomplete ATM Transactions



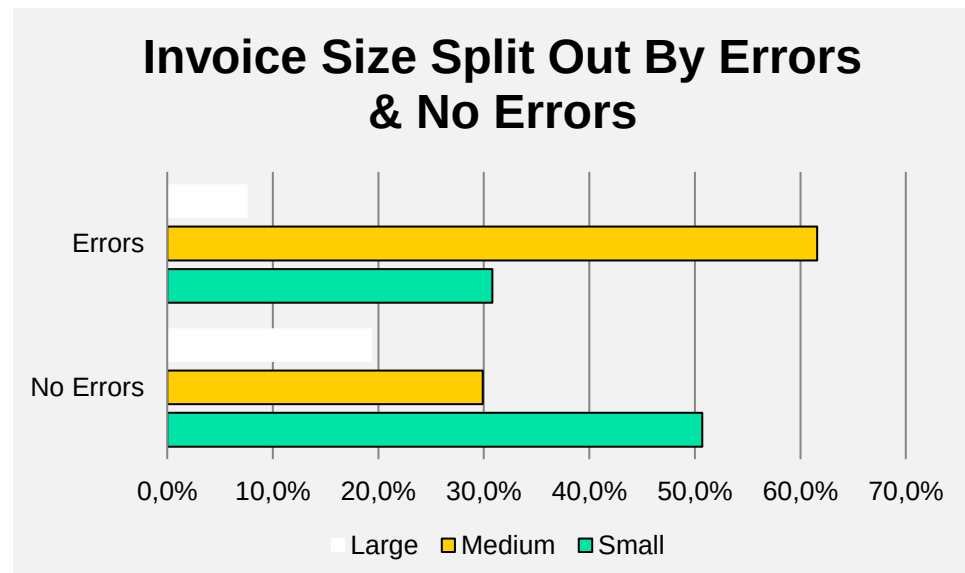
The “Vital Few”

Visualising Categorical Data: Side-By-Side Bar Charts

DCOVA

- The **side-by-side bar chart** represents the data from a contingency table.

	No Errors	Errors	Total
Small Amount	50.75%	30.77%	47.50%
Medium Amount	29.85%	61.54%	35.00%
Large Amount	19.40%	7.69%	17.50%
Total	100.0%	100.0%	100.0%



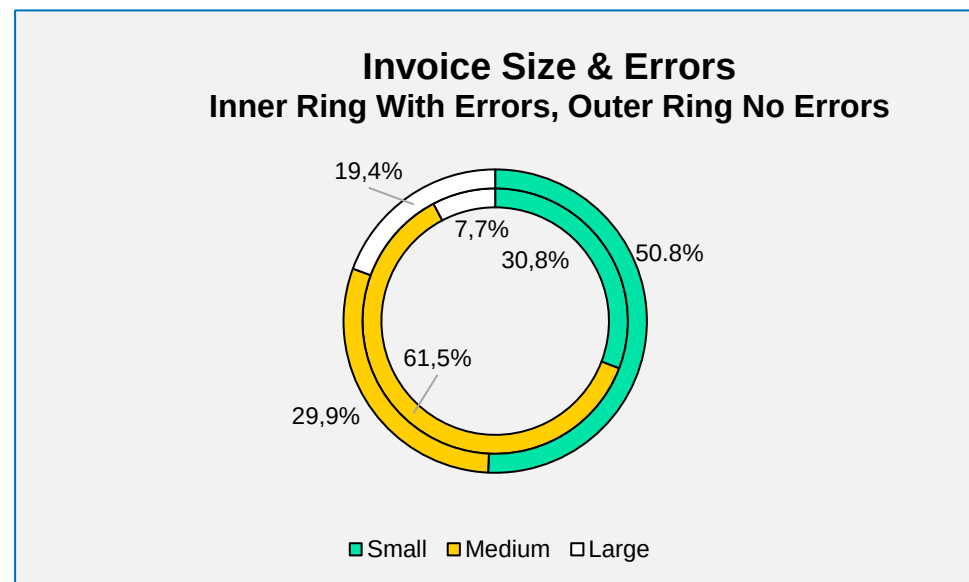
Invoices with errors are much more likely to be of medium size (61.5% vs 30.8% & 7.7%).

Visualising Categorical Data: Doughnut Charts

DCOVA

- A **Doughnut Chart** can be used to represent the data from a contingency table.

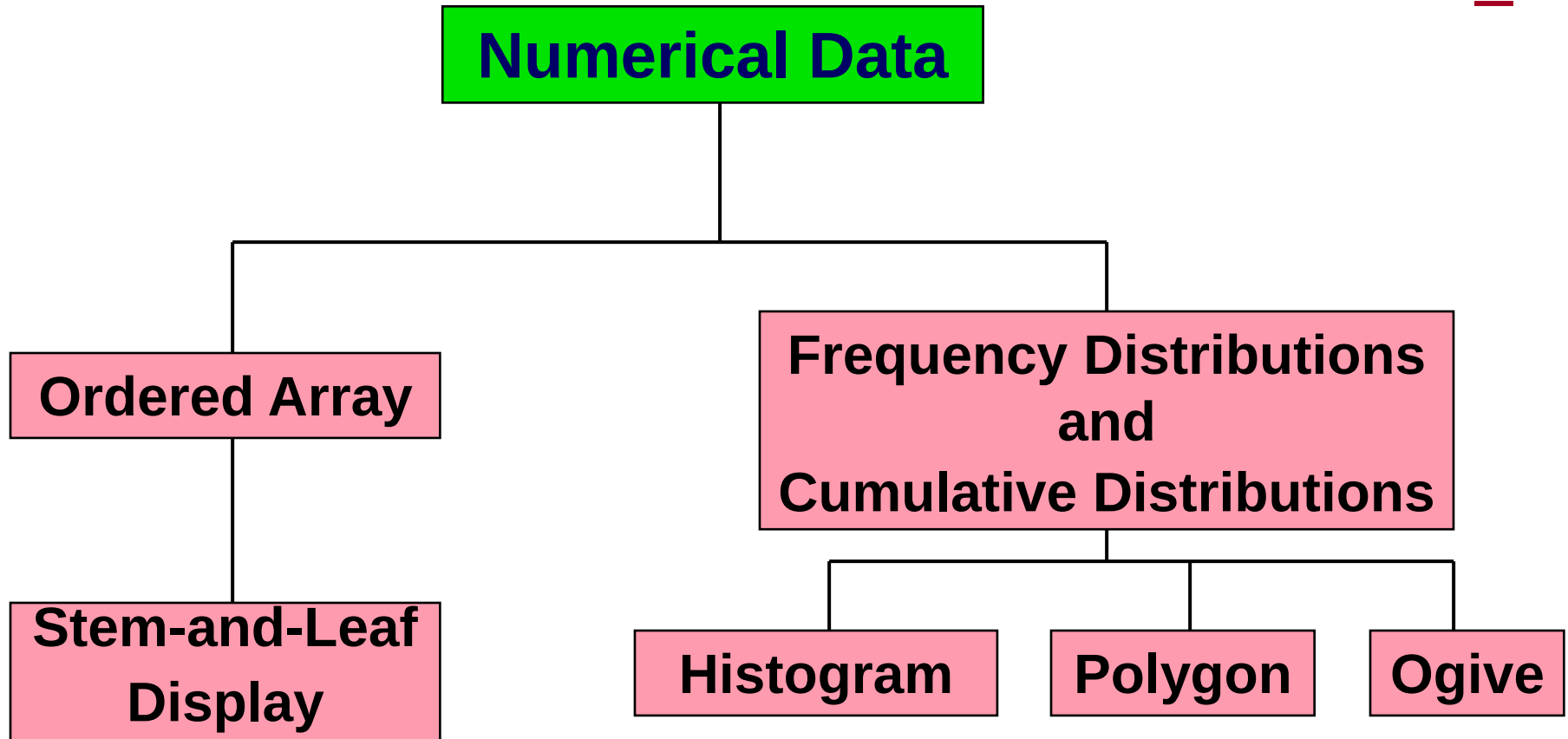
	No Errors	Errors	Total
Small Amount	50.75%	30.77%	47.50%
Medium Amount	29.85%	61.54%	35.00%
Large Amount	19.40%	7.69%	17.50%
Total	100.0%	100.0%	100.0%



Invoices with errors are much more likely to be of medium size (61.5% vs 30.8% & 7.7%).

Visualising Numerical Data By Using Graphical Displays

DCOVA



Stem-and-Leaf Display

DCOVA

- A simple way to see how the data are distributed and where concentrations of data exist.

METHOD: Separate the sorted data series into leading digits (the **stems**) and the trailing digits (the **leaves**).

Organising Numerical Data: Stem and Leaf Display

DCOVA

- A **stem-and-leaf display** organises data into groups (called stems) so that the values within each group (the leaves) branch out to the right on each row.

Age of College Students

Age of Surveyed College Students	Day Students					
	16	17	17	18	18	18
	19	19	20	20	21	22
	22	25	27	32	38	42
	Night Students					
	18	18	19	19	20	21
	23	28	32	33	41	45

Day Students

Stem	Leaf
1	67788899
2	0012257
3	28
4	2

Night Students

Stem	Leaf
1	8899
2	0138
3	23
4	15

Visualising Numerical Data: The Histogram

DCOVA

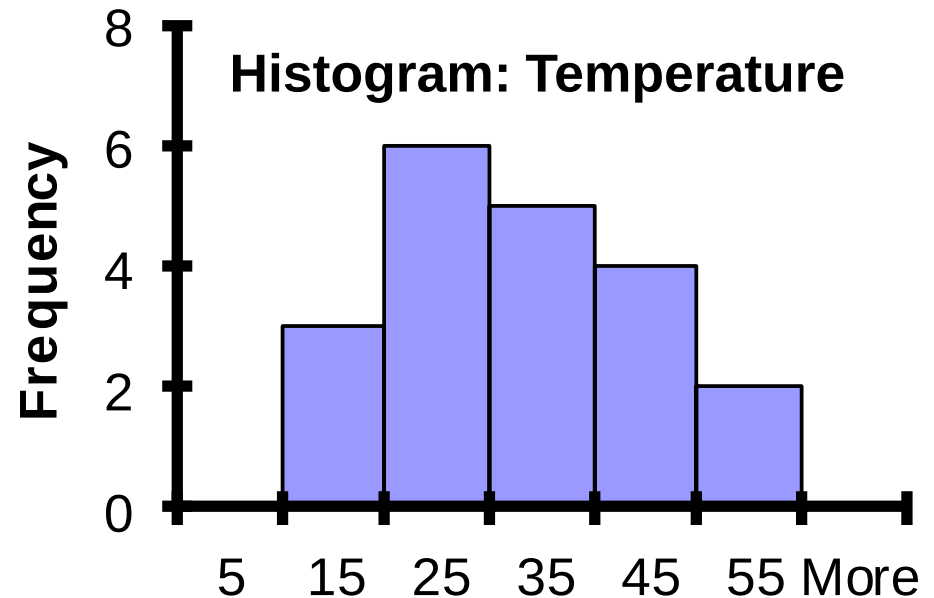
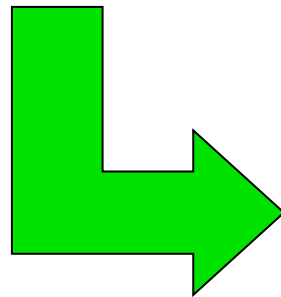
- A vertical bar chart of the data in a frequency distribution is called a **histogram**.
- In a histogram there are no gaps between adjacent bars.
- The **class boundaries** (or **class midpoints**) are shown on the horizontal axis.
- The vertical axis is either **frequency**, **relative frequency**, or **percentage**.
- The height of the bars represent the frequency, relative frequency, or percentage.

Visualising Numerical Data: The Histogram

DCOVA

Class	Frequency	Relative Frequency	Percentage
10 but less than 20	3	.15	15
20 but less than 30	6	.30	30
30 but less than 40	5	.25	25
40 but less than 50	4	.20	20
50 but less than 60	2	.10	10
Total	20	1.00	100

(In a percentage histogram the vertical axis would be defined to show the percentage of observations per class).



Visualising Numerical Data: The Percentage Polygon

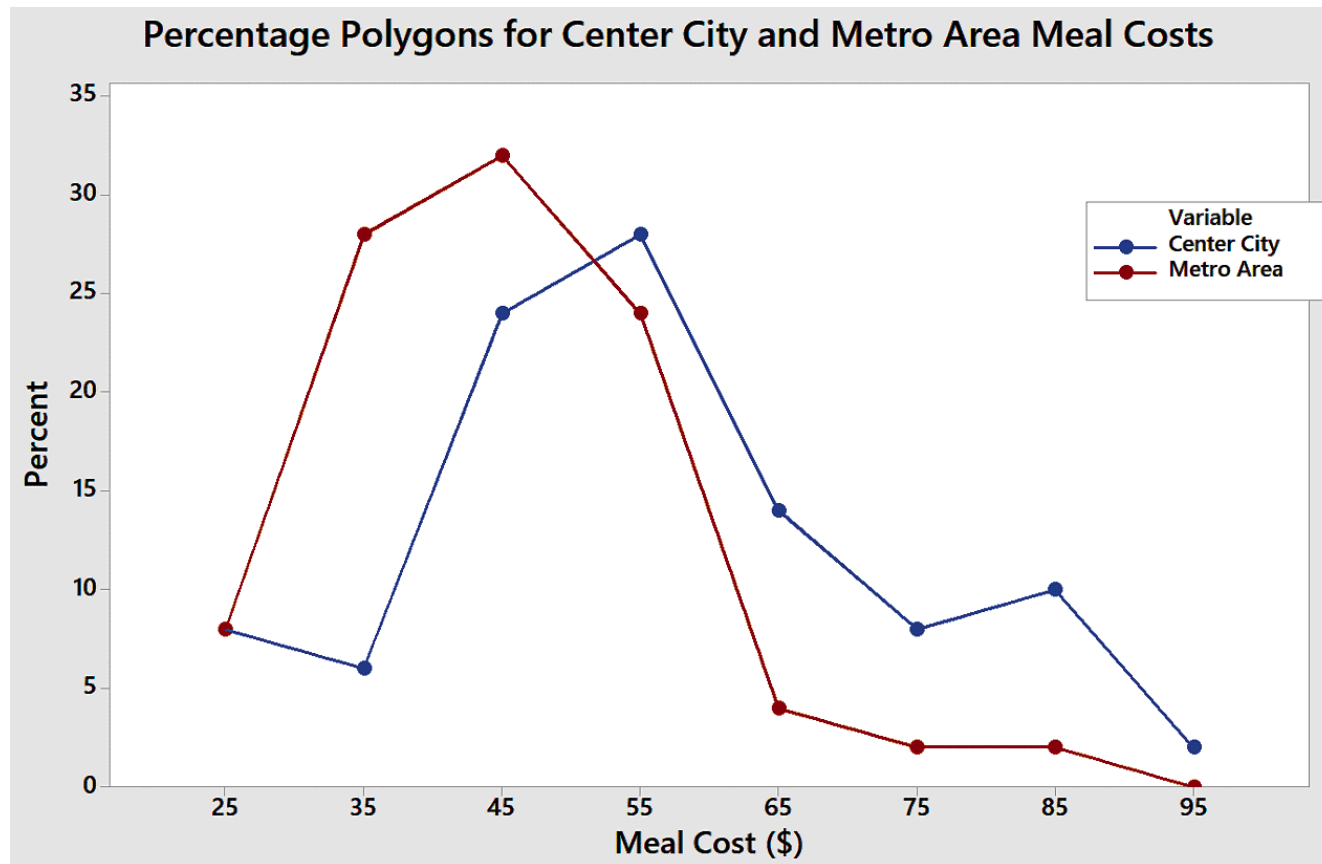
DCOVA

- A **percentage polygon** is formed by having the midpoint of each class represent the data in that class and then connecting the sequence of midpoints at their respective class percentages.
- The **cumulative percentage polygon**, or **ogive**, displays the variable of interest along the X axis, and the cumulative percentages along the Y axis.
- Useful when there are two or more groups to compare.

Visualising Numerical Data: The Percentage Polygon

DCOVA

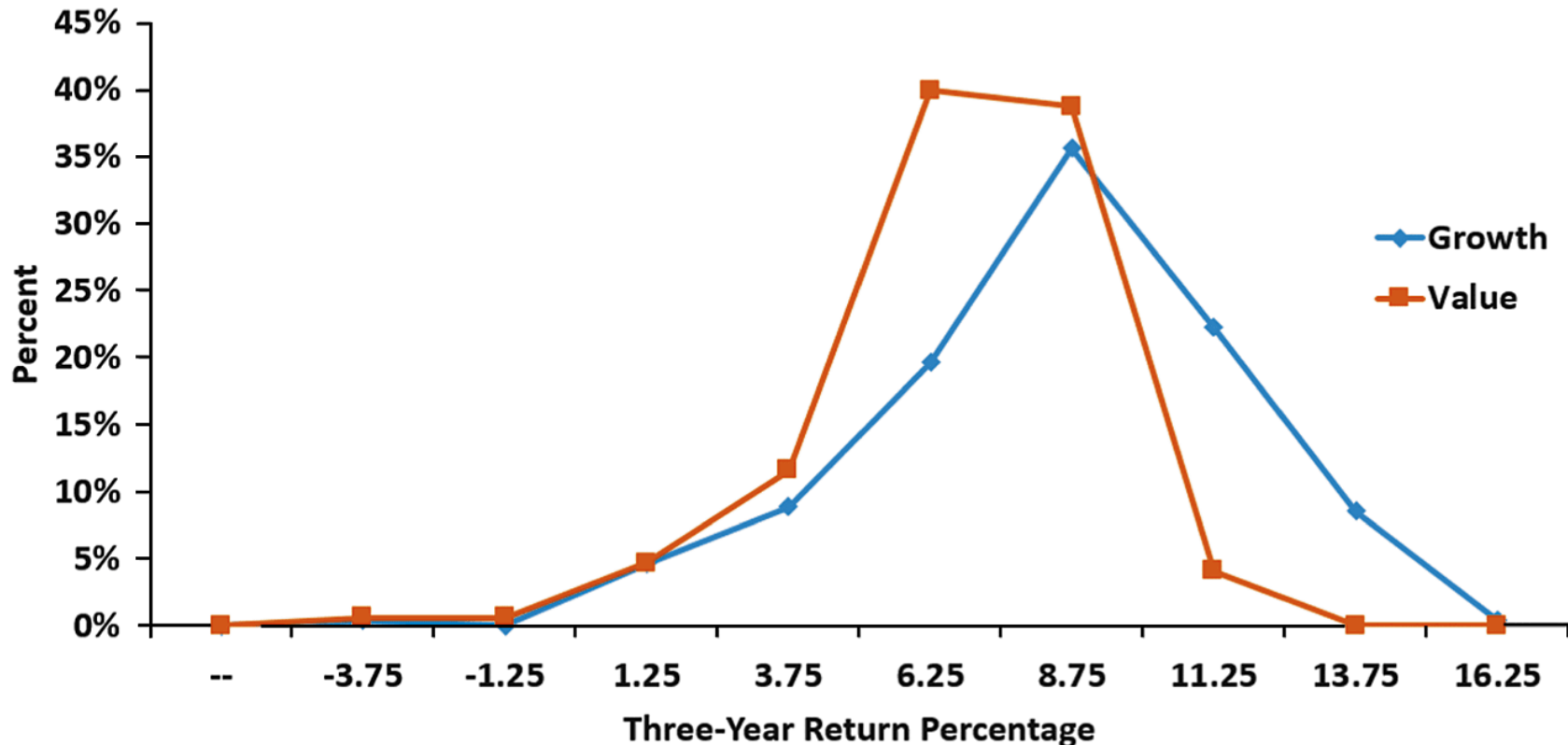
Useful When Comparing Two or More Groups



Visualising Numerical Data: The Percentage Polygon

DCOVA

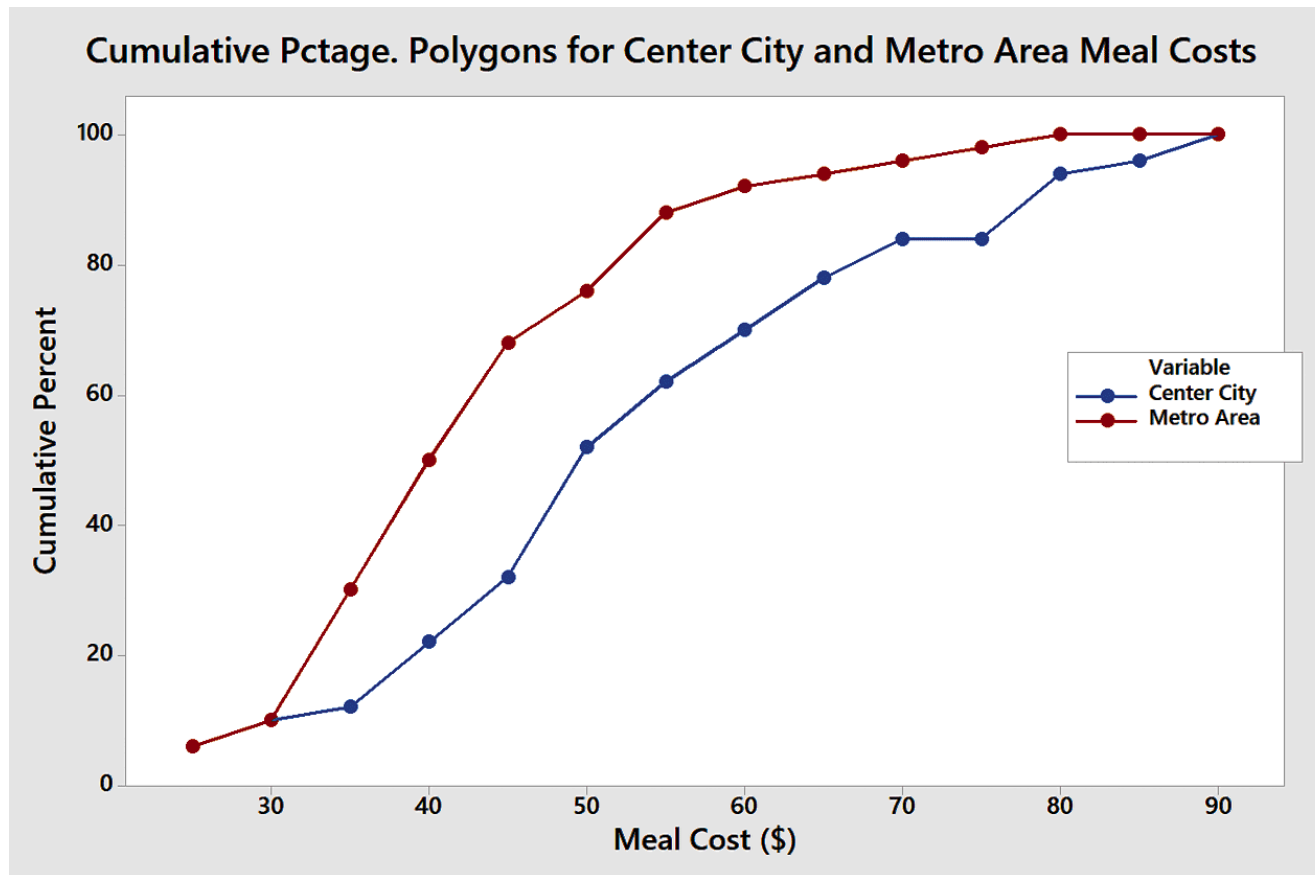
Percentage Polygons for Three-Year Return Percentage
for the Growth and Value Funds



Visualising Numerical Data: The Cumulative Percentage Polygon (Ogive)

DCOVA

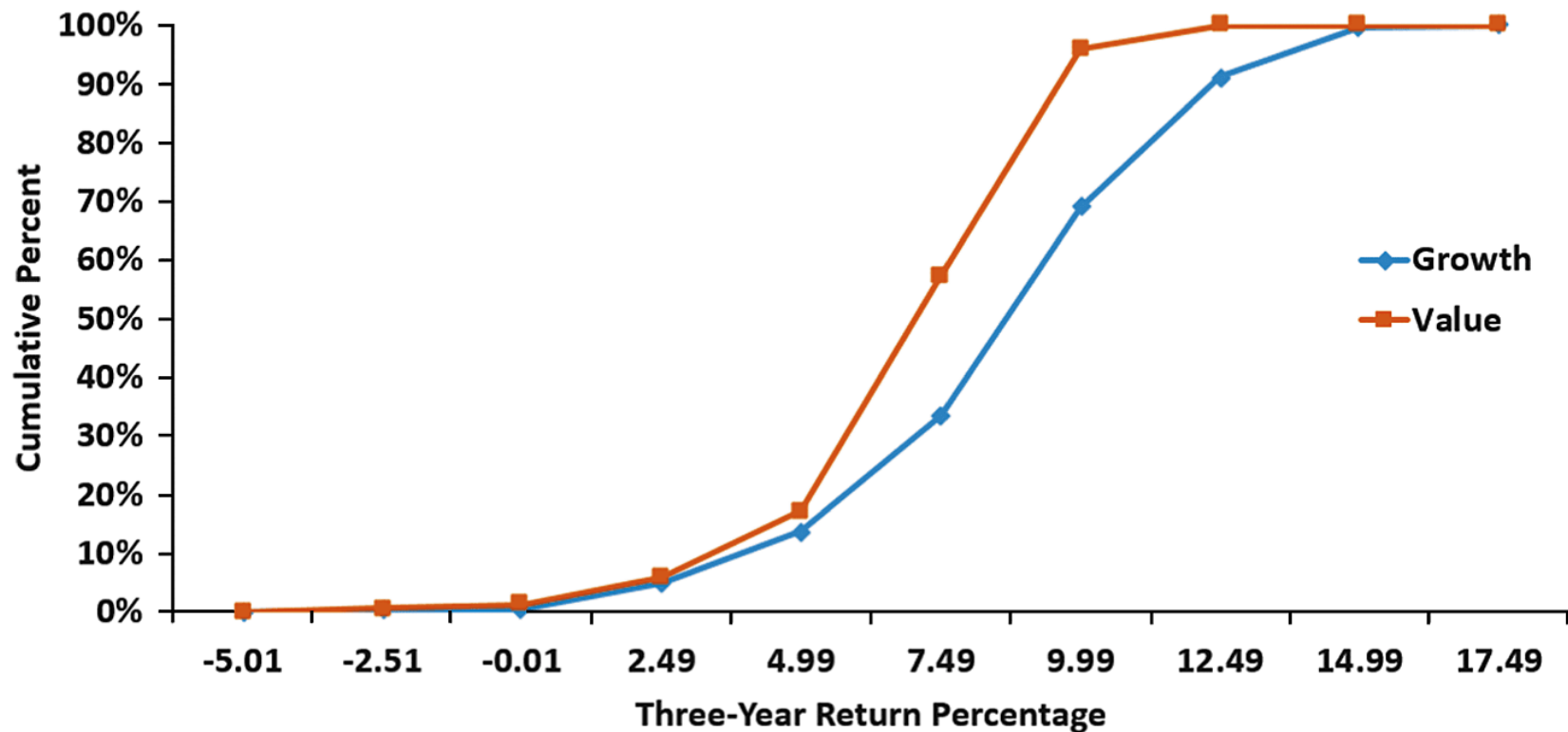
Useful When Comparing Two or More Groups



Visualising Numerical Data: The Cumulative Percentage Polygon (Ogive)

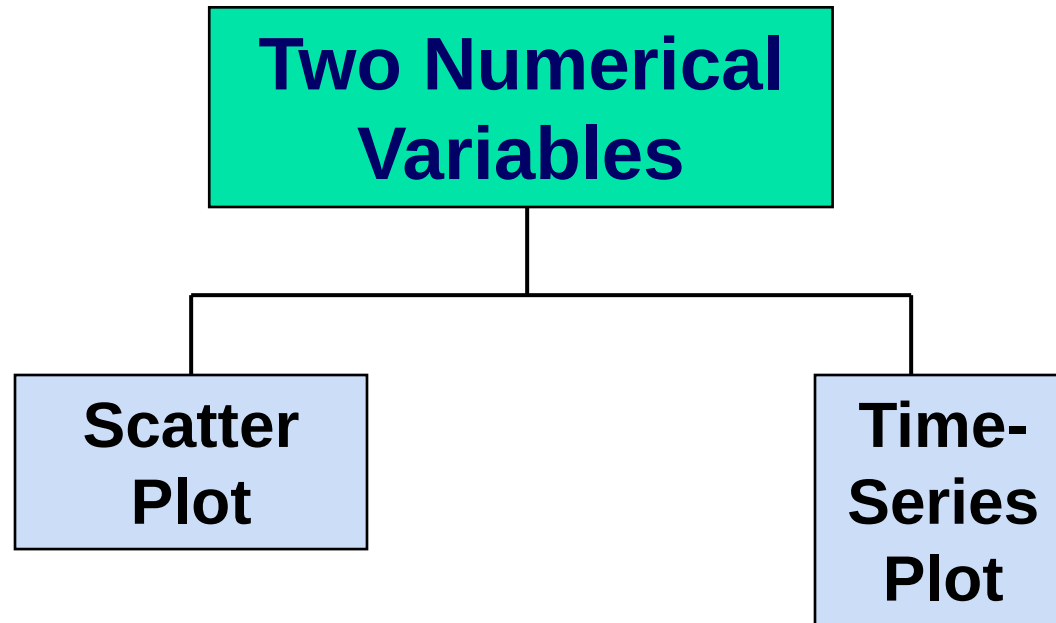
DCOVA

Cumulative Percentage Polygons for the
Three-Year Return Percentages for the Growth and Value Funds



Visualising Two Numerical Variables By Using Graphical Displays

DCOVA



Visualising Two Numerical Variables: The Scatter Plot

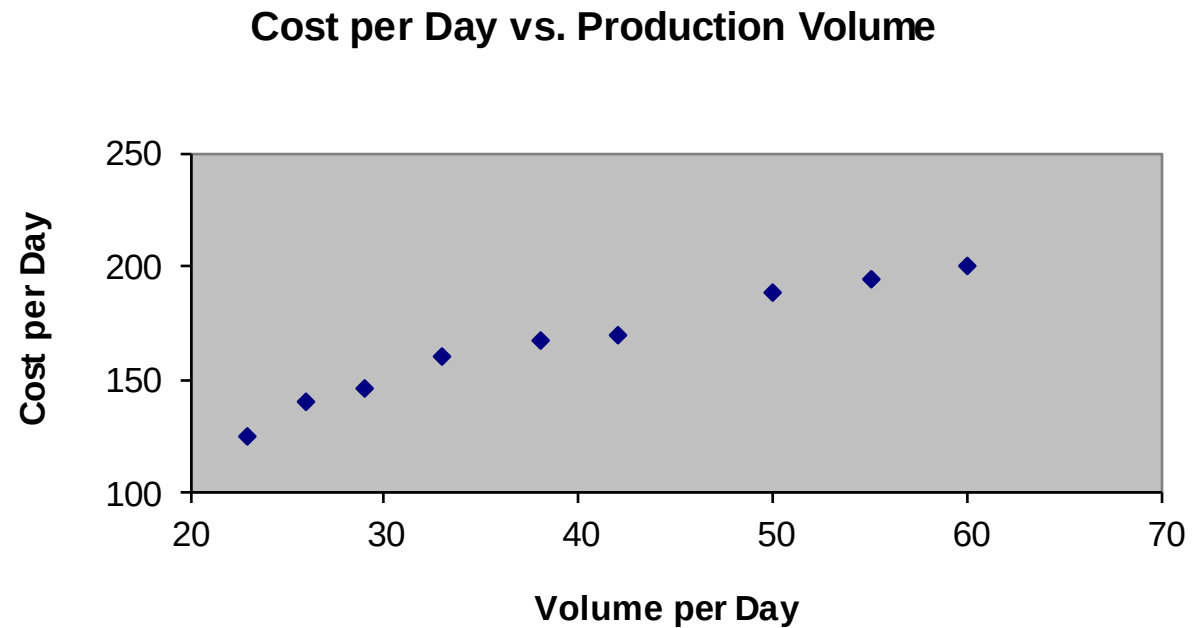
DCOVA

- **Scatter plots** are used for numerical data consisting of paired observations taken from two numerical variables.
- One variable's values are displayed on the horizontal or X axis and the other variable's values are displayed on the vertical or Y axis.
- Scatter plots are used to examine possible relationships between two numerical variables.

Scatter Plot Example

DCOVA

Volume per day	Cost per day
23	125
26	140
29	146
33	160
38	167
42	170
50	188
55	195
60	200



Visualising Two Numerical Variables: The Time Series Plot

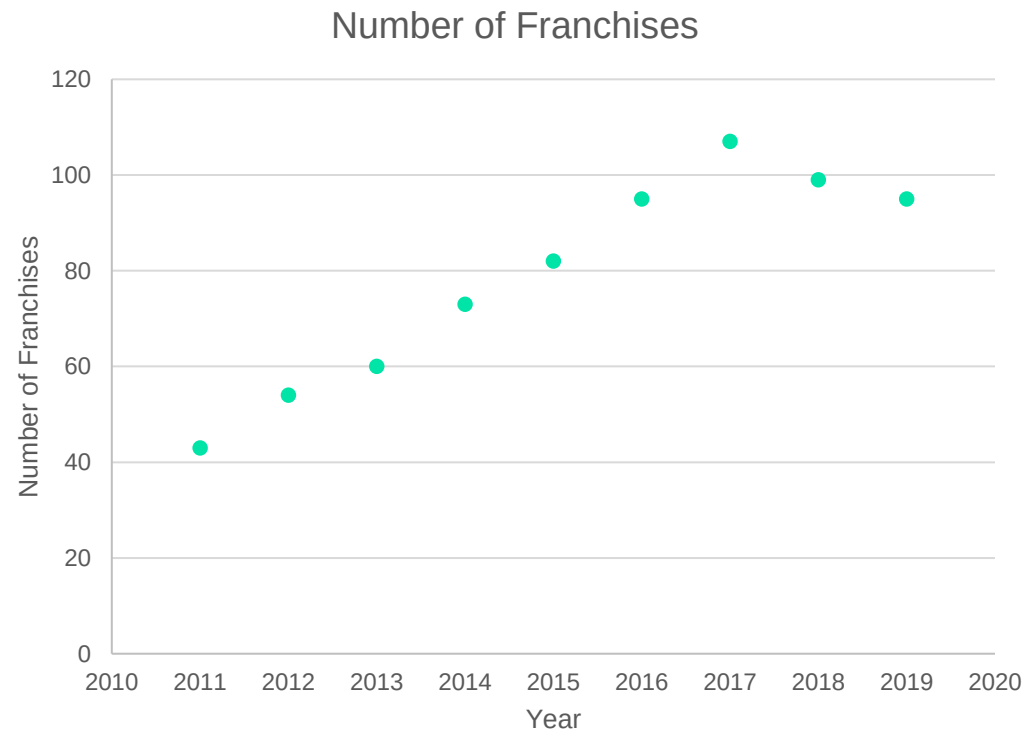
DCOVA

- A **Time-Series Plot** is used to study patterns in the values of a numeric variable over time.
- The Time-Series Plot:
 - Numeric variable's values are on the vertical axis and the time period is on the horizontal axis.

Time Series Plot Example

DCOVA

Year	Number of Franchises
2011	43
2012	54
2013	60
2014	73
2015	82
2016	95
2017	107
2018	99
2019	95



Organising A Mix Of Variables: The Multidimensional Contingency Table

DCOVA

- A **multidimensional contingency table** is constructed by tallying the responses of three or more categorical variables.
 - Can be used to discover possible patterns and relationships in multidimensional data that simpler tables and charts would fail to make apparent.
- As a practical rule, tables should be limited to no more than three or four variables.
- In typical use, these tables:
 - Extend contingency tables to two or more row or column variables, or
 - Replace the frequencies found in a contingency table with summary information about a numeric variable.

A Multidimensional Contingency Table Tallies Responses Of Three or More Categorical Variables

DCOVAA

Two Dimensional Table Showing Fund Type and Risk Level for sample of 479 retirement funds.

Risk Level v				
Fund Type v	Low	Average	High	Grand Total
Growth	13.15%	31.73%	19.00%	63.88%
Value	17.54%	15.03%	3.55%	36.12%
Grand Total	30.69%	46.76%	22.55%	100.00%

Three Dimensional Table Showing Fund Type, Market Cap, and Risk Level for the sample of the 479 retirement funds.

Risk Level v				
Fund Type v	Low	Average	High	Grand Total
☒ Growth	13.2%	31.7%	19.0%	63.88%
Large	9.6%	19.0%	3.5%	32.2%
MidCap	3.3%	9.4%	5.2%	18.0%
Small	0.2%	3.3%	10.2%	13.8%
☒ Value	17.5%	15.0%	3.5%	36.1%
Large	14.6%	7.9%	0.6%	23.2%
MidCap	2.1%	3.5%	0.8%	6.5%
Small	0.8%	3.5%	2.1%	6.5%
Grand Total	30.69%	46.76%	22.55%	100.00%

In Excel creating a Pivot Table yields an interactive display of this type.

Drilling-Down On A Table Reveals The Data The Table Summarizes

DCOVA

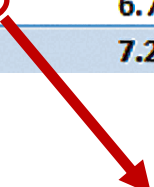
- Clicking a cell in an Excel table displays the rows of data associated with that cell.
- Drill-down is perhaps the simplest form of data discovery.

Drill-Down Reveals The Data Underlying A Higher-Level Summary

DCOVA

Mean 10YrReturn	Risk Level 			
Fund Type 	Average	High	Low	Grand Total
 Growth	7.78	7.19	8.06	7.66
Large	7.91	7.88	8.04	7.94
MidCap	7.41	6.60	8.10	7.30
Small	8.14	7.25	8.47	7.49
 Value	6.52	5.97	6.45	6.43
Large	5.87	4.18	6.30	6.10
MidCap	7.69	6.15	6.99	7.27
Small	6.79	6.43	7.61	6.78
Grand Total	7.38	7.00	7.14	7.22

Results of drilling down to the details about small value funds with low risk revealing the ten-year return ranges from 4.83% to 9.44%.




	F	G	H	I	J	K	L	M	N	O
1	Turnover Ratio 	SD 	Sharpe Ratio 	Beta 	1YrReturn 	3YrReturn 	5YrReturn 	10YrReturn 	Expense Ratio 	Star Rating 
2	75.00	4.67	0.53	0.20	4.74	2.53	4.82	9.44	1.40	Four
3	23.00	9.61	0.72	0.84	7.74	6.95	8.17	7.30	1.28	Four
4	30.10	10.71	0.70	0.55	5.88	7.60	7.74	4.83	1.70	Two
5	37.00	11.73	0.71	0.79	6.02	8.45	9.29	8.85	0.81	Four

Displays To Visualise A Mix Of Many Variables

DCOVA

- Displays are more useful than a multidimensional contingency table with many row and column variables.
- The data (not just summary statistics) can be shown for numerical variables.
- Multiple numerical variables can be presented in one summarisation.
- Visualisations can reveal patterns that can be hard to see in tables.

Colored Scatter Plots Visualise Both Numerical Variables & Categorical Variable(s)

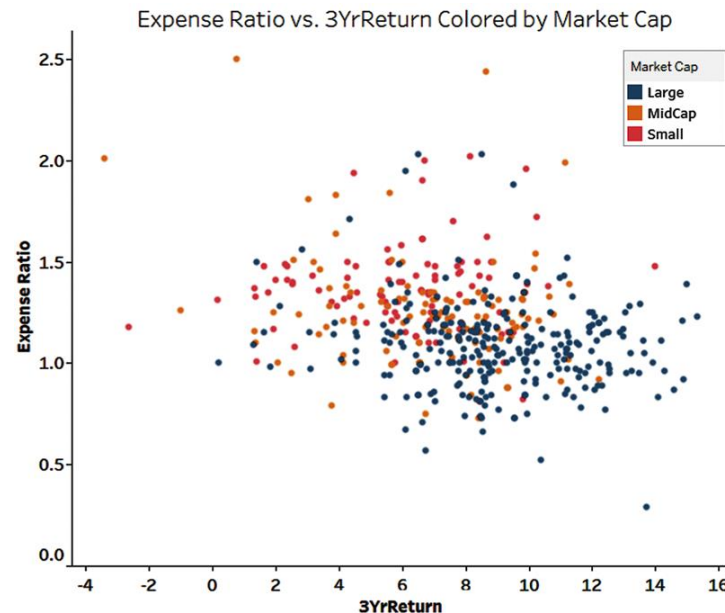
Observations:

Large Market Capitalization Funds (blue dots)

DCOVA

1. Relatively have best returns and lowest expense ratios.
2. Some have either low returns or high expense ratios or both.

Tableau Scatter Plot



Bubble Charts Extend Scatter Plots

DCOVA

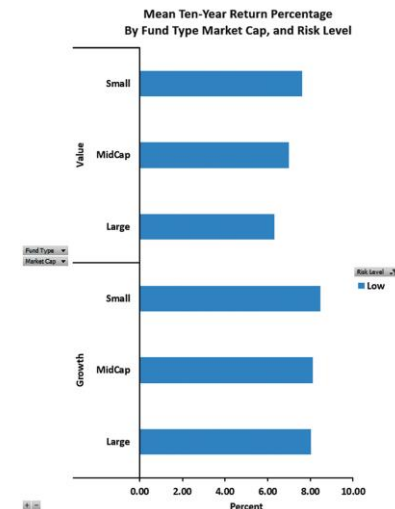
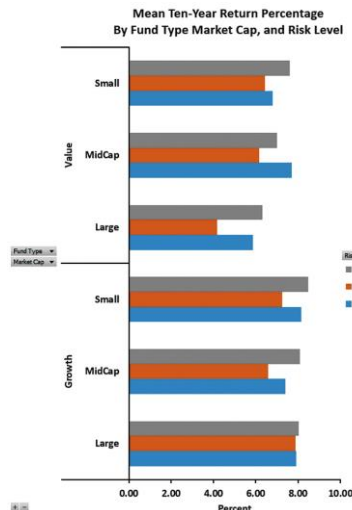
- Use the size of points (called bubbles) to represent the value of an additional variable.
- In Excel the additional variable must be numerical.
- In Tableau the additional variable can be either numerical or categorical.

An Excel PivotChart Visualises Specific Categories From A PivotTable Summary

DCOVA

Mean 10YrReturn	Risk Level ↑↓			
Fund Type ▼	Average	High	Low	Grand Total
Growth	7.78	7.19	8.06	7.66
Large	7.91	7.88	8.04	7.94
MidCap	7.41	6.60	8.10	7.30
Small	8.14	7.25	8.47	7.49
Value	6.52	5.97	6.45	6.43
Large	5.87	4.18	6.30	6.10
MidCap	7.69	6.15	6.99	7.27
Small	6.79	6.43	7.61	6.78
Grand Total	7.38	7.00	7.14	7.22

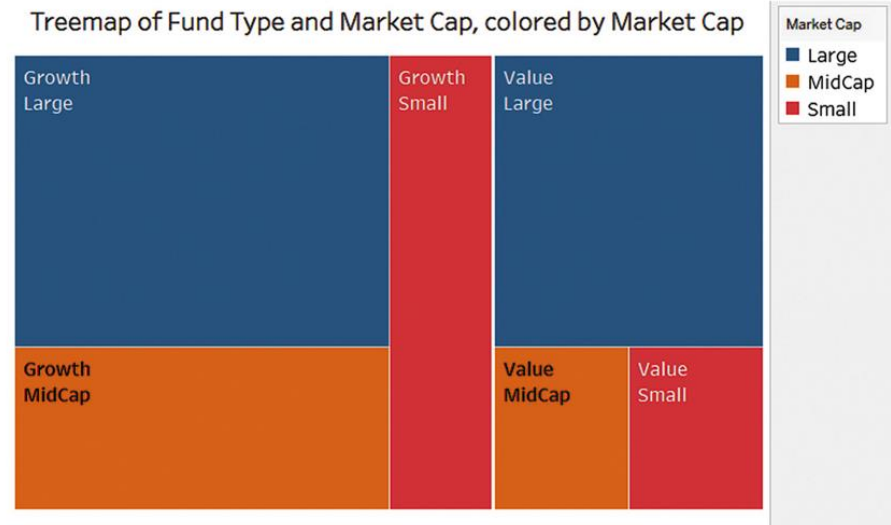
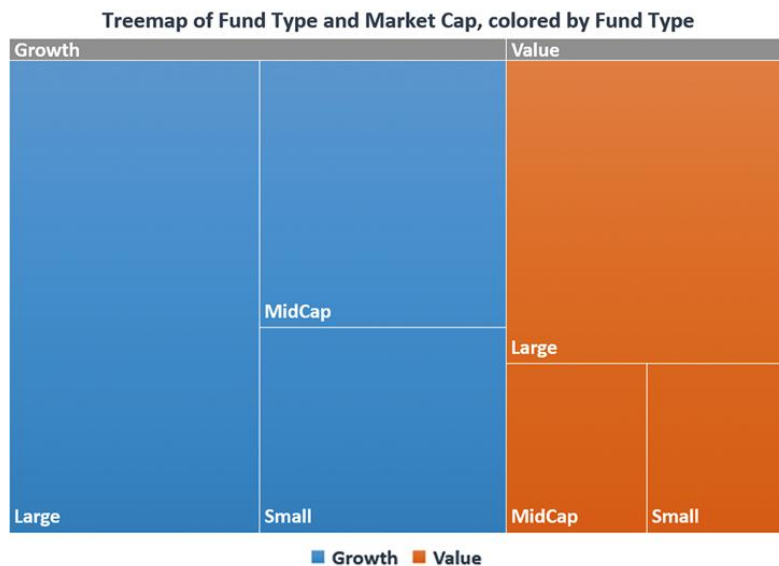
**Low Risk Small
Market Cap Funds
Have The Highest Mean
Return Among Low Risk
Funds.**



Treemaps Are Graphical Displays Of Contingency Tables

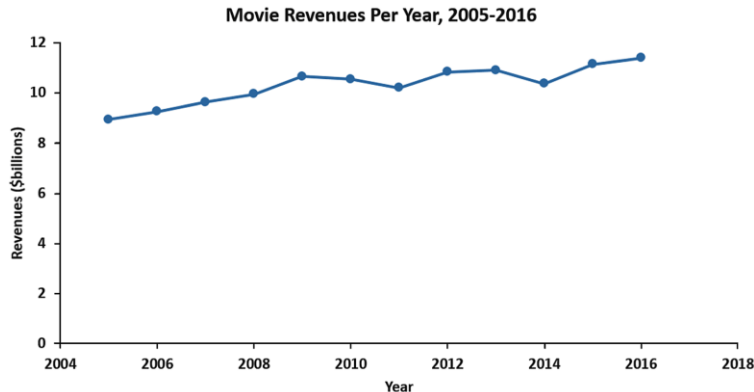
DCOVA

Excel and Tableau Treemaps For Fund Type and Market Cap



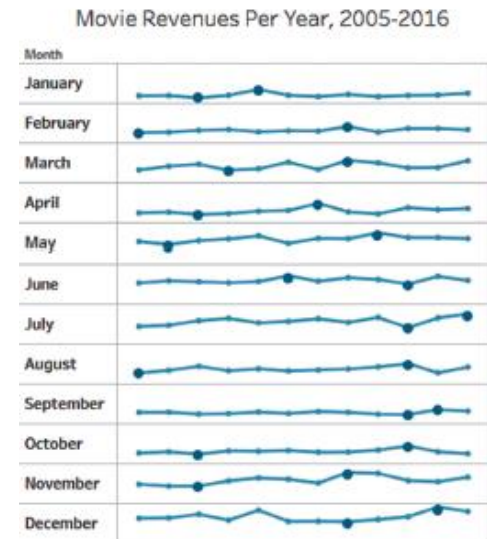
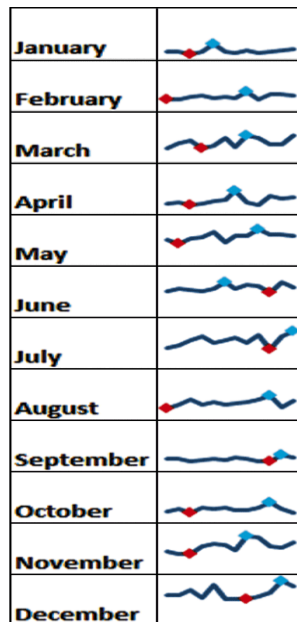
Sparklines Are Compact Time-Series Visualisations Of Numerical Variables

DCOVA



Movie
revenues
by week
per month

Excel and
Tableau
Sparklines
Displays



Filtering and Querying Data

DCOVA

- Two operations associated with preparing tabular or visual summaries are Data Filtering and Querying.
- Data Filtering selects rows of data that match criteria; specified value(s) for specific variable(s).
- Data Querying is similar but may not select all the columns of the matching rows.
- Excel has various filtering & query features.

Excel Slicers Filter & Query Data From A Pivot Table

DCOVA

- A slicer is a panel of clickable buttons superimposed over a worksheet.
- Each button in a slicer represents a unique value of a variable found in the source data of a PivotTable.
- By clicking buttons in the slicer panels, you query the data.

Example Of Slicers For The Retirement Funds Workbook

DCOVA

With the four slicers below, you can ask questions such as:

1. What are the attributes of the fund(s) with the lowest expense ratio?
2. What are the expense ratios associated with large market cap value funds that have a star rating of five?

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	PivotTable of Fund Type and Risk Level													
2														
3		Risk Level ▾												
4	Fund Type ▾	Low	Average	High	Grand Total									
5	Growth	63	152	91	306									
6	Value	84	72	17	173									
7	Grand Total	147	224	108	479									
8														
9														
10														
11														
12														
13														
14														

Fund Type

Growth

Value

Star Rating

Five

Four

One

Three

Two

Market Cap

Large

MidCap

Small

Expense Ratio

0.29

0.52

0.57

0.66

0.67

0.71

0.73

0.74

Answering The Questions

DCOVA

What are the attributes of the fund(s) with the lowest expense ratio?

What are the expense ratios associated with large market cap value funds that have a star rating of five?

The updated PivotTable (not shown below) reveals only one such fund.

The expense ratios for these funds are: 0.83, 0.94, 1.05, 1.09, 1.18, and 1.19.

Fund Type	Star Rating	Expense Ratio
Growth	Five	0.29
Value	Four	0.52
	One	0.57
	Three	0.66
	Two	0.67
		0.71
		0.73
		0.74

Fund Type	Star Rating	Expense Ratio
Growth	Five	0.83
Value	Four	0.94
	One	1.05
	Three	1.09
	Two	1.18
		1.19
		0.29
		0.52

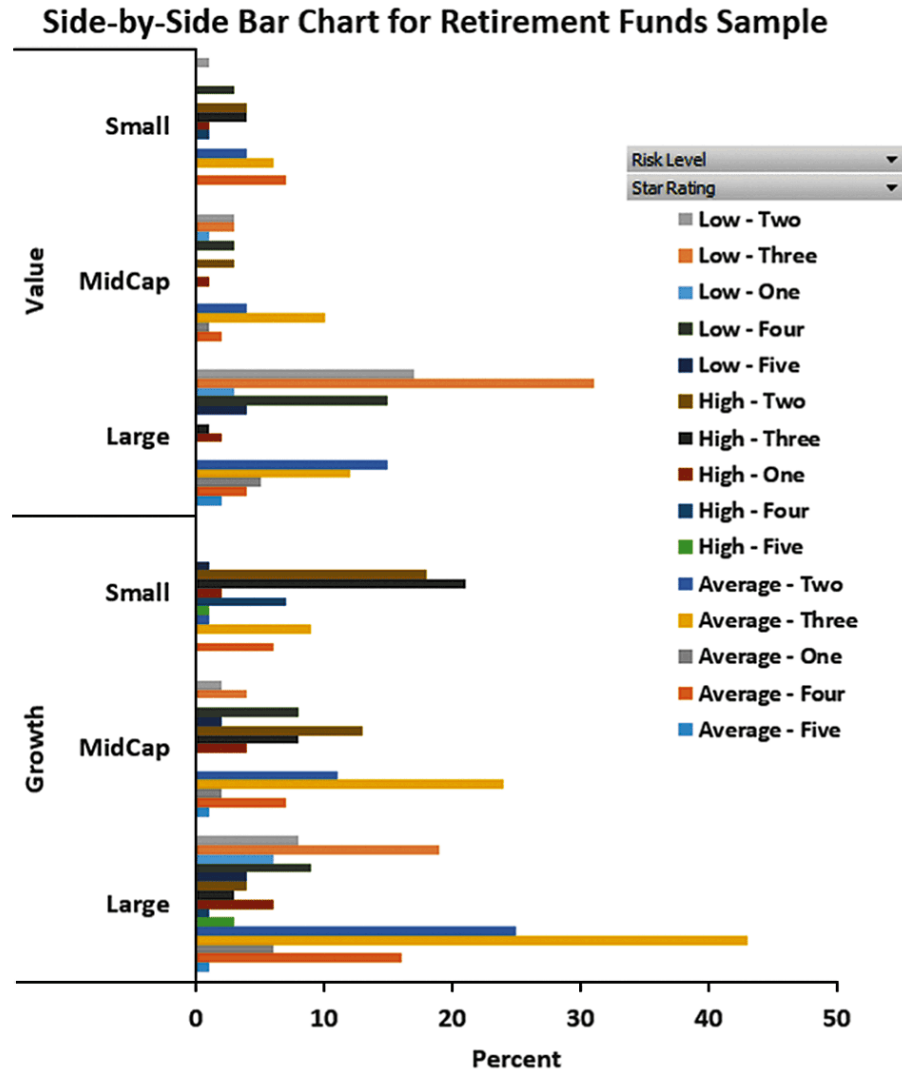
The Pitfalls in Organising and Visualising Variables

DCOVA

- When organising and visualising data need to be mindful of:
 - The limits of other's ability to perceive and comprehend.
 - Presentation issues that can undercut the usefulness of methods from this chapter.
- It is easy to create summaries that:
 - Obscure the data or
 - Create false impressions.
 - Contain Chartjunk

An Example Of Obscuring Data, Information Overload

DCOVAA



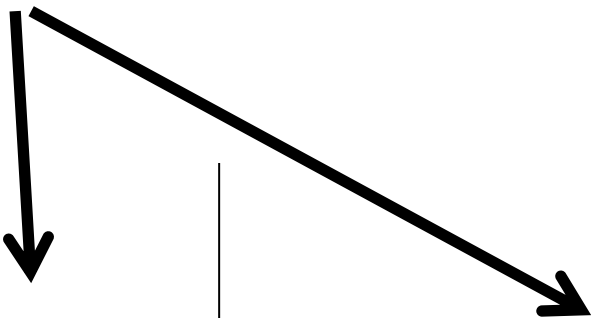
False Impressions Can Be Created In Many Ways

DCOVA

- Selective summarisation:
 - Presenting only part of the data collected.
- Improperly constructed charts:
 - Potential pie chart issues.
 - Improperly scaled axes.
 - A Y axis that does not begin at the origin or is a broken axis missing intermediate values.
- Chartjunk.

An Example of Selective summarisation, These Two Summarisations Tell Totally Different Stories

DCOVA

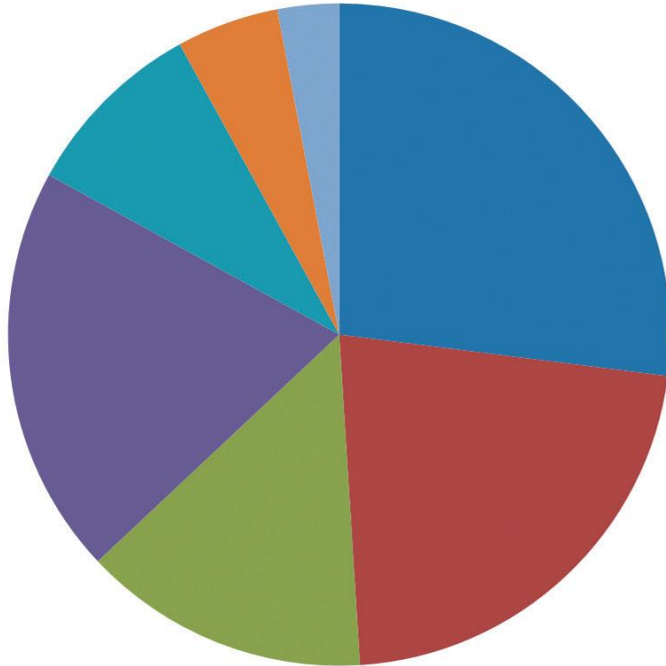


Company	Change from Prior Year	Company	Year 1	Year 2	Year 3
A	+7.2%	A	-22.6%	-33.2%	+7.2%
B	+24.4%	B	-4.5%	-41.9%	+24.4%
C	+24.9%	C	-18.5%	-31.5%	+24.9%
D	+24.8%	D	-29.4%	-48.1%	+24.8%
E	+12.5%	E	-1.9%	-25.3%	+12.5%
F	+35.1%	F	-1.6%	-37.8%	+35.1%
G	+29.7%	G	+7.4%	-13.6%	+29.7%

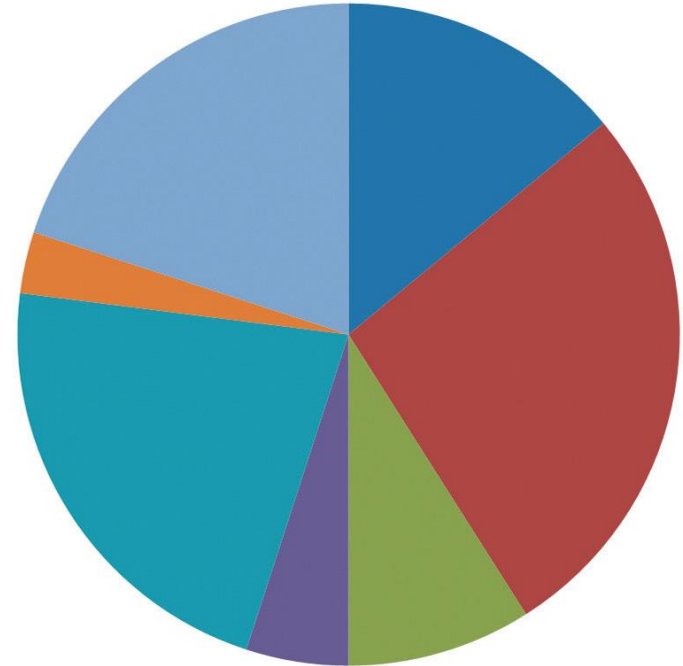
How Obvious Is It That Both Pie Charts Summarize The Same Data?

DCOVA

Market Share of Companies



Market Share of Companies



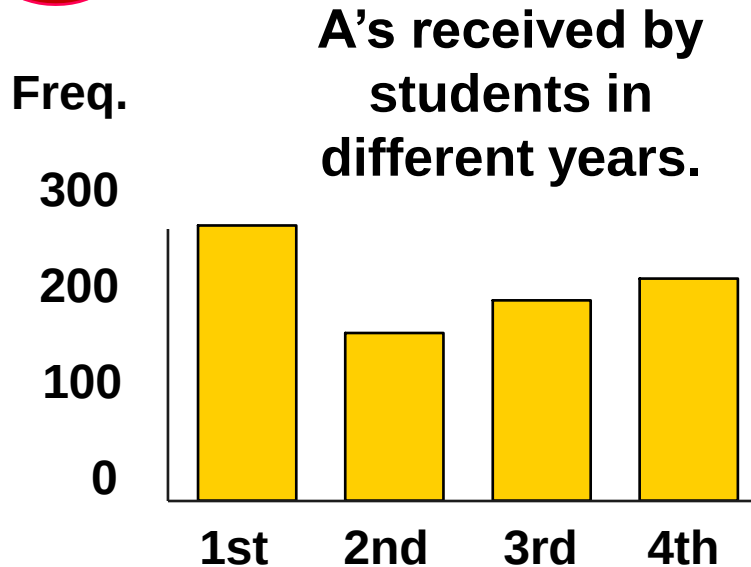
Why is it hard to tell? What would you do to improve?

Graphical Errors: No Relative Basis

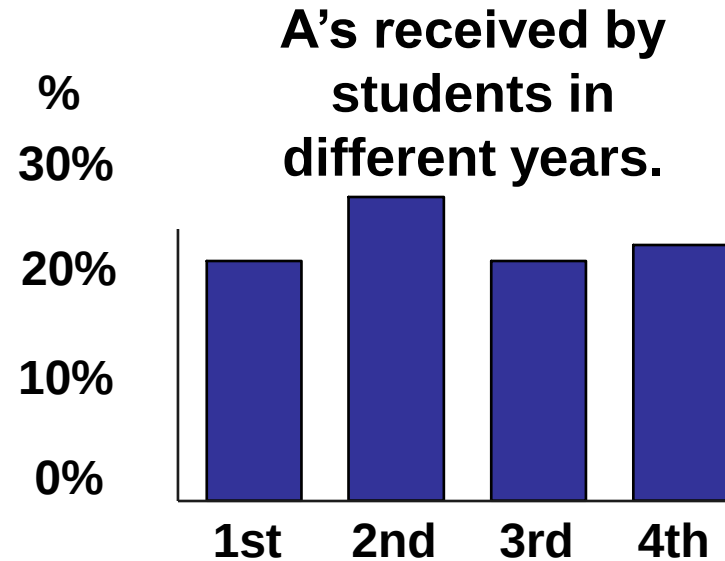
DCOVA



Bad Presentation



Good Presentation

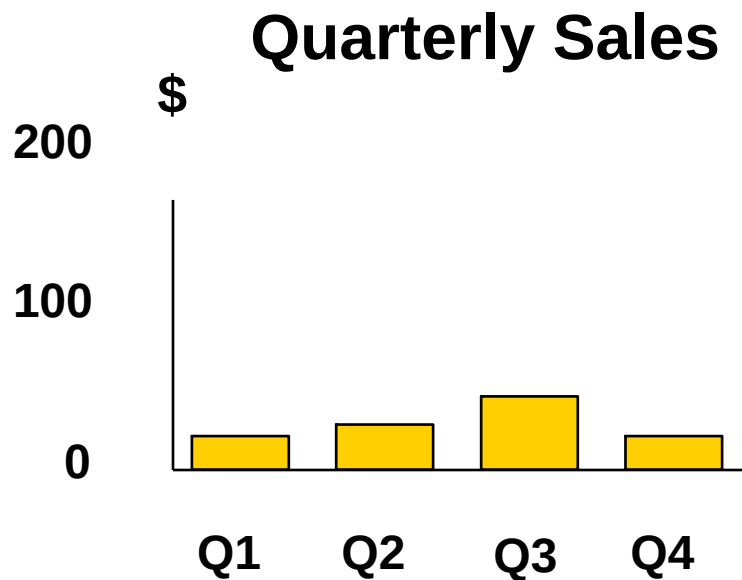


Graphical Errors: Compressing the Vertical Axis

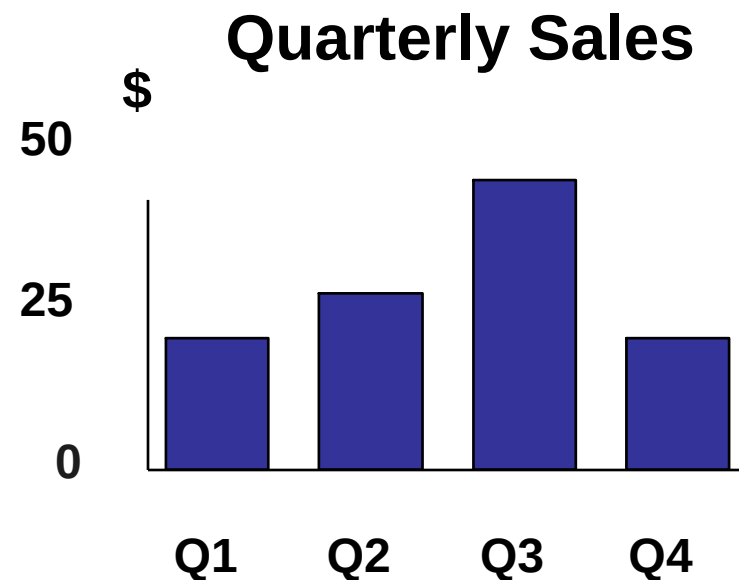
DCOVA



Bad Presentation

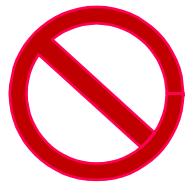


Good Presentation

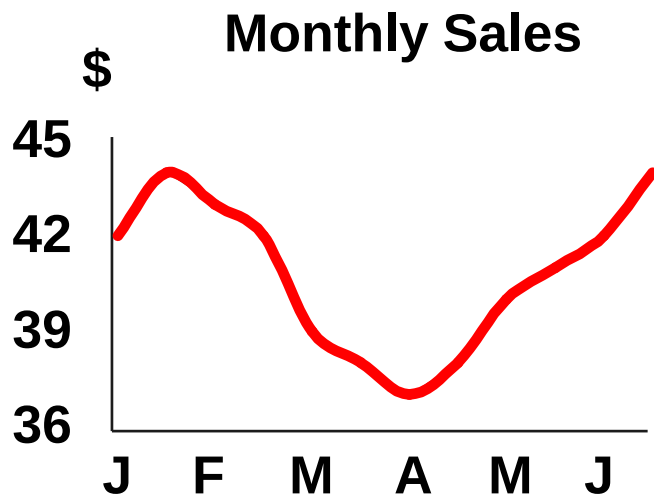


Graphical Errors: No Zero Point on the Vertical Axis

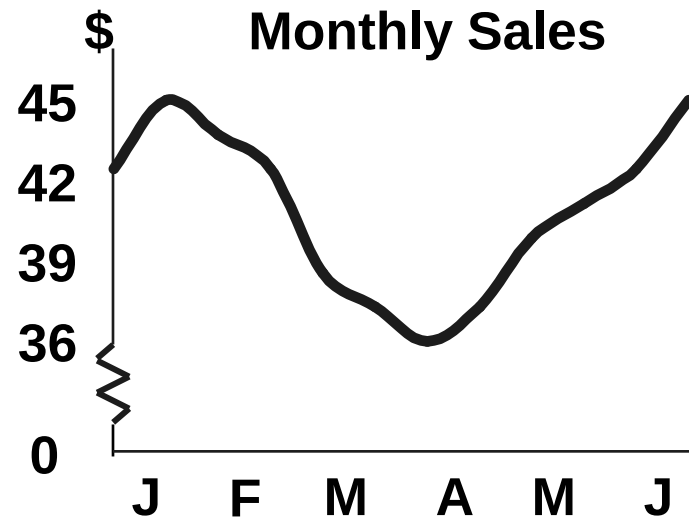
DCOVA



Bad Presentation



✓ Good Presentations



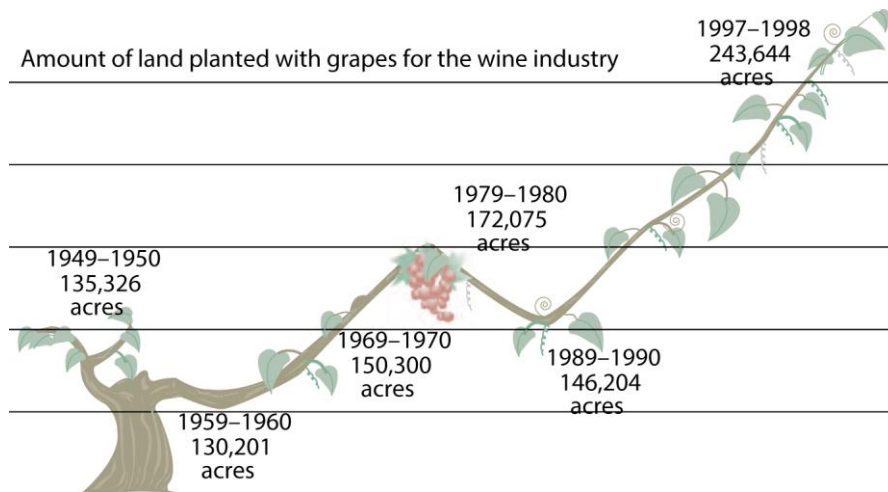
Graphing the first six months of sales

Graphical Errors: Chart Junk, Can You Identify The Junk?

DCOVA



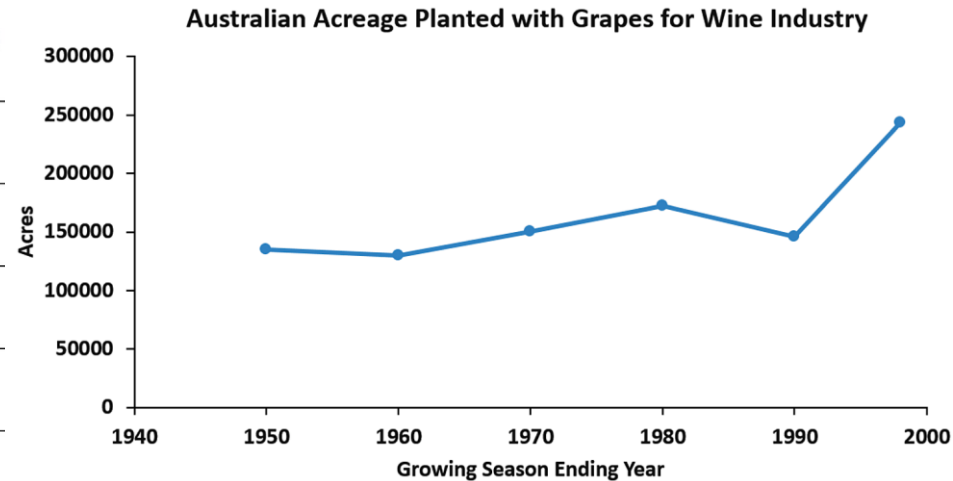
Bad Presentation



Left illustration adapted from S. Watterson, "Liquid Gold—Australians Are Changing the World of Wine. Even the French Seem Grateful." Time, November 22, 1999, p. 68-69



Good Presentation



Best Practices for Constructing Visualisations

DCOVA

- Use the simplest possible visualization.
- Include a title & label all axes.
- Include a scale for each axis if the chart contains axes.
- Begin the scale for a vertical axis at zero & use a constant scale.
- Avoid 3D or “exploded” effects & the use of chartjunk.
- Use consistent colorings in charts meant to be compared.
- Avoid using uncommon chart types including radar, surface, cone, and pyramid charts.

Chapter Summary

In this chapter we covered:

- Organising and visualising categorical variables.
- Organising and visualising numerical variables.
- Summarizing a mix of variables.
- Avoiding common errors when organising and visualising variables.